EXAMPLE 8

Computation of Returns

Ulli Lohrmann and his wife, Suzanne Lohrmann, are planning for retirement and want to compare the past performance of a few mutual funds they are considering for investment. They believe that a comparison over a five-year period would be appropriate. They gather information on a fund they are considering, the Rhein Valley Superior Fund, which is presented in Exhibit 12.

Exhibit 12: Rhein Valley Superior Fund Performance

Year	Assets under Management at the Beginning of Year (euros)	Annual Return (%)
1	30 million	15
2	45 million	−5
3	20 million	10
4	25 million	15
5	35 million	3

The Lohrmanns are interested in aggregating this information for ease of comparison with other funds.

Exhibit 13: Rhein Valley Superior Fund Annual Returns and Investments (euro millions)

Year	1	2	3	4	5
Balance from previous year	0	34.50	42.75	22.00	28.75
New investment by the investor (cash inflow for the Rhein fund)	30.00	10.50	0	3.00	6.25
Withdrawal by the investor (cash outflow for the Rhein fund)	0	0	−22.75	0	0
Net balance at the beginning of year	30.00	45.00	20.00	25.00	35.00
Investment return for the year	15%	−5%	10%	15%	3%
Investment gain (loss)	4.50	−2.25	2.00	3.75	1.05
Balance at the end of year	34.50	42.75	22.00	28.75	36.05

1. Compute the fund's holding period return for the five-year period.

   Solution:

   The five-year holding period return is calculated as:

   $$R = (1 + \mathrm{R}_1)(1 + \mathrm{R}_2)(1 + \mathrm{R}_3)(1 + \mathrm{R}_4)(1 + \mathrm{R}_5) - 1$$

   $$R = (1.15)(0.95)(1.10)(1.15)(1.03) - 1 =$$

   $$R = 0.4235 = 42.35\%.$$

2. Compute the fund' s arithmetic mean annual return.

   Solution:

   The arithmetic mean annual return is calculated as: